

Aravali International School, Sector 43

Project Documentation & Proposal

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Institution: Aravali International School, Sector 43, Faridabad

Date: September 27, 2026

TASK 1: AI Research Report

Title: Implementation of AI-Powered Smart Library System

1. Introduction

Traditional school library management often struggles with manual tracking, inefficient book recommendations, and mismanaged cataloging. The proposed **AI-Powered Smart Library System** utilizes artificial intelligence, optical character recognition (OCR), and machine learning algorithms to automate book issuing, track inventories, and personalize reading recommendations for students at Aravali International School, Sector 43.

2. Objectives

- To automate book cataloging, check-in, and check-out processes using AI vision systems.
- To provide personalized reading suggestions to students based on their borrowing history and interests.
- To optimize resource allocation and inventory management in real time.

3. Benefits of the AI Solution

- Time Savings:** Reduces student wait times during peak library hours by up to 70%.
- Personalized Learning:** Enhances student reading engagement through tailored recommendation models.
- Accuracy:** Minimizes human error in record-keeping, tracking, and overdue notifications.

4. Key Features

- Smart Cataloging:** Automated cover and ISBN scanning using OCR.
- Predictive Recommendations:** AI engine providing customized book lists to students.
- Automated Overdue Alerts:** AI-scheduled reminders sent directly to students and parents.

5. Resources Required

- Hardware:** High-definition webcams, RFID scanners, and server hardware.
- Software:** Python (OpenCV, Scikit-Learn), MySQL Database, and cloud hosting infrastructure.
- Personnel:** School IT administrator, librarian trainer, and project supervisor.

TASK 2: AI Content Creation

Aravali International School, Sector 43

NOTICE

Date: September 27, 2026

Subject: Launch of AI-Powered Smart Library Project

This is to inform all students and staff that Aravali International School, Sector 43 will be introducing the **AI-Powered Smart Library System** starting next month. This system aims to make book searches and issuance seamless, fast, and personalized. All students are requested to update their library profiles in the IT lab by October 15, 2026.

Principal,

Aravali International School, Sector 43

EVENT ANNOUNCEMENT

AI Tech Expo 2026: Innovations in Education

Date: November 10, 2026

Time: 9:00 AM – 3:00 PM

Venue: School Main Auditorium, Sector 43 Campus

Join us for the unveiling of our students' AI projects! Experience live demonstrations of the new **Smart Library System**, dynamic chatbots, and intelligent tutoring solutions developed by our AI Club members.

AI PROJECT DESCRIPTION

Project Name: AI Smart Library Assistant

Overview: The Smart Library Assistant utilizes natural language processing (NLP) and recommendation algorithms to catalog books and assist students in finding relevant academic materials.

Target Audience: K–12 Students, Teachers, and Library Staff.

Expected Impact: Streamlining library management and fostering a stronger reading culture across all grade levels.

TASK 3: AI Promotional Poster Layout Guide



ज़रूरी जानकारी
क्लास 9 के लिए



ARAVALI
INTERNATIONAL SCHOOL

READING · LIBRARY SYSTEM: THE FUTURE OF LEARNING:

THE SMART LIBRARY SYSTEM



1. INSTANT
BOOK DISCOVERY



2. AUTOMATED
CHECKOUTS



3. TAILORED READS
TECHNOLOGIES



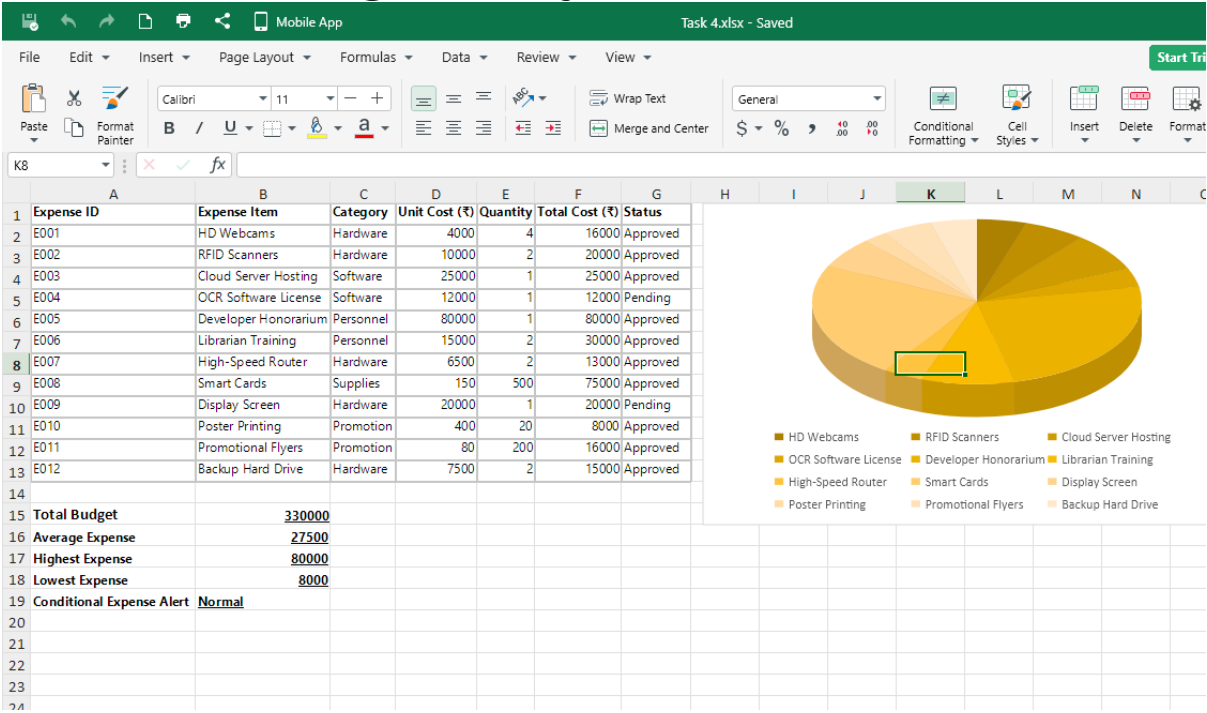
QR CODE

(SCAN TO EXPLORE)

DATE: NOVEMBER 10, 2026

VENUE: • MAIN AUDITORIUM

TASK 4: AI Budget Analysis



TASK 5: AI Participant Database

Participants

	A	B	C	D	E	
1	ParticipantID	FirstName	LastName	Grade	Email	
2	101	Mayank	Lal	Class 9	mayank@example.com	
3	102	Aarnav	Verma	Class 9	aarnav@example.com	
4	103	Rohan	Mehta	Class 10	rohan@example.com	
5	104	Ananya	Gupta	Class 9	ananya@example.com	
6	105	Priyansh	Singh	Class 8	priyansh@example.com	
7						

Events

	A	B	C	D
1	EventID	EventName	EventDate	ParticipantID
2	E1	AI Tech Expo 2026	11-10-2026	101
3	E2	AI Tech Expo 2026	11-10-2026	102
4	E3	AI Tech Expo 2026	11-10-2026	103
5	E4	Smart Library Workshop	2026-10-15	101
6	E5	Smart Library Workshop	2026-10-15	104

Query

	A	B	C	D	E	F
1	ParticipantID	Event ID	Event Name	First Name	Last Name	Grade
2	101	E1	AI Tech Expo 2026	Mayank	Lal	Class 9
3	102	E1	AI Tech Expo 2026	Aarnav	Verma	Class 9

Form

	A	B	C	D	E
1	Row	Column A	Column B	Column C	Column D
2	PARTICIPANT REGISTRATION FORM				
3	Participant ID	106	(User Input Field)		
4	First Name	Aarav	(User Input Field)		
5	Last Name	Kapoor	(User Input Field)		
6	Grade	Class 9	(User Input Field)		
7	Email	aarav@example.com	(User Input Field)		
8	SUBMIT ENTRY				

TASK 6: AI Presentation Structure

Slide 1:

Aravali International School, Sector 43

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AI-POWERED SMART LIBRARY SYSTEM

Intelligent Automation for Modern School Libraries



Presenter: Lavish Sharma

September 27, 2026

Slide 2:

Aravali International School, Sector 43

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INTRODUCTION & OBJECTIVES

School libraries are critical to student learning, yet most institutions still rely on manual, paper-based processes for book management, checkouts, and record-keeping. These outdated workflows lead to inefficiencies, errors, and reduced engagement. The AI-Powered Smart Library System addresses this challenge by automating school library workflows using Artificial Intelligence — enabling faster operations, smarter recommendations, and a significantly reduced administrative burden on library staff.



Streamline Book Allocation

Automate the process of issuing, returning, and tracking books to eliminate manual errors and reduce wait times for students and staff.



Dynamic Recommendation Models

Implement AI-driven recommendation algorithms that suggest personalized reading materials to students based on their reading history and preferences.



Reduce Manual Administrative Load

Minimize repetitive administrative tasks such as manual cataloguing, record updates, and inventory checks through intelligent automation and smart reporting.

Slide 3:

AI SOLUTION & KEY FEATURES

Our solution integrates OCR vision systems with advanced Machine Learning algorithms to automate and enhance every aspect of the school library experience – from checkout to inventory management.



AUTOMATED CHECKOUT

OCR scanner enables fast, error-free book checkout without manual input.



Reading Recommendations

Personalized ML models suggest books based on student interests and history.



Smart Inventory

AI-driven inventory tracking reduces loss and optimizes book availability.

Slide 4:

BUDGET ANALYSIS & BENEFITS

Estimated total project expense has been calculated across three key categories: Hardware, Software, and Setup. The AI-Powered Smart Library System delivers measurable value through fast checkouts, increased student engagement, and reduced manual record usage.



Hardware Costs

45%

Scanners, RFID tags, kiosks, and computing devices required for automated checkout and smart inventory management infrastructure.

Software & AI

35%

Machine learning models, OCR vision systems, recommendation engine licenses, and library management software integration costs.

Setup & Training

20%

Installation, system configuration, staff training, and initial onboarding to ensure smooth deployment and adoption across the library.

Slide 5:

CONCLUSION

Future Scope: The AI-Powered Smart Library System can be further enhanced by integrating voice assistant support for hands-free access and facial recognition technology for secure, seamless student identification – improving accessibility and automating library entry and checkout processes.

Conclusion: AI has the transformative potential to fully modernize school library operations. By automating checkouts, personalizing recommendations, and reducing manual workloads, this system sets a new standard for smart, efficient, and student-centered library management at Aravali International School.



Slide 6:

THANK YOU!



Questions & Discussion



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Aravali International School



Sector 43, Faridabad



Github Link: <https://github.com/slavishpro-spec/CT-AI-Project-Submission.git>